Practical 8 - Yacc/Bison Code

Yacc file: practcc8.y

Open practcc8.y

```
/*Name:- Mohammed Zaid Shaikh
Prn:- 24070521043
Sec:- A2
*/

%{
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int yylex();
void yyerror(char *errorMsg);
%}

%token NUMBER

%%
stmt: expr '\n' {printf("Result: %d\n", $1); }

expr: NUMBER {$$ = $1;}
    | expr expr '+' {$$ = $1 + $2;}
    | expr expr '-' {$$ = $1 - $2;}
    | expr expr '*' {$$ = $1 * $2;}
    | expr expr '/' {$$ = $1 / $2;}
    | expr expr '%' {$$ = $1 % $2;}
    | expr expr '^' {$$ = (int)pow($1, $2);}
%%

void yyerror(char *errorMsg) {
    printf("Error");
    exit(0);
}

int main() {
    yyparse();
    return 0;
}
```

Practical 8 - Lex Code and Output

Lex file: practcc8.l

Open practcc8.l

```
/*
Name:- Mohammed Zaid Shaikh
Prn:- 24070521043
Sec:- A2
*/

%{
#include <stdio.h>
#include "y.tab.h"
%}

%%
[0-9]+ { yylval = atoi(yytext); return NUMBER; }
[+*-^%/] { return yytext[0]; }
[ \t] { ; }
\n { return '\n'; }
. { printf("Error"); }
%%

int yywrap() {
    return 1;
}
```

Output

```
zaid787@LAPTOP-ID0Q783M:~$ gedit practcc8.y
zaid787@LAPTOP-ID0Q783M:~$ gedit practcc8.l
zaid787@LAPTOP-ID0Q783M:~$ yacc -d practcc8.y
zaid787@LAPTOP-ID0Q783M:~$ flex practcc8.l
zaid787@LAPTOP-ID0Q783M:~$ gcc y.tab.c lex.yy.c -lm
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
2 5 ^
Result: 32
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
25 3 %
Result: 1
2 5 ^ 3 *
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
05 3 % 16 +
Result: 18
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
10 6 3 * 4 - 2 / +
Result: 17
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
8 2 * 7 +
Result: 23
```